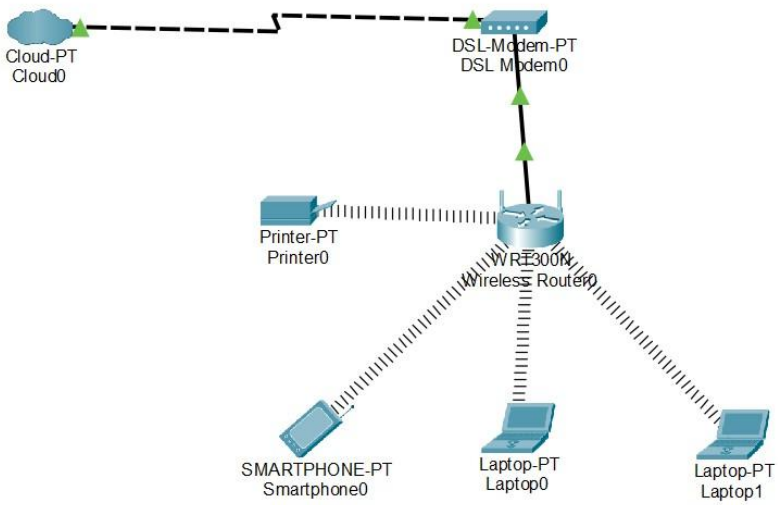
BUILD SOHO NETWORK

1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

Ans:-



Connection Details:

1. Broadband Internet connects to the Wi-Fi router using a wired connection.
2. Laptop 0 connects to the router using Wi-Fi (Wireless).
3. Laptop 1 connects to the router using Wi-Fi (Wireless).
4. Smart Phone connects to the router using Wi-Fi (Wireless).
5. Printer connects to the router using an Ethernet cable (Wired).

Wireless (Wi-Fi) Connections:

1. Wireless Router → Printer0
2. Wireless Router → Smartphone0
3. Wireless Router → Laptop0
4. Wireless Router → Laptop1

2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Ans:-

No.	Hardware Component	Brand / Model Suggestion	Qty.	Approx. Price (₹)
1	Broadband Modem / ONT	JioFiber / Airtel ONT	1	1,500–3,000
2	Wi-Fi Router	TP-Link Archer AX23	1	4,000–5,500
3	Laptop	Dell / HP Laptop	1	35,000–45,000
4	Smart TV	Samsung / LG Smart TV (32-inch)	1	12,000–18,000
5	Ethernet Cable	CAT6 D-Link / FEDUS	2	200–500
6	RJ45 Connectors	D-Link / FEDUS	4	100–300
7	Wi-Fi Range Extender	TP-Link RE315	1	2,500–3,500
8	Network Switch	TP-Link TL-SG1005D (5-Port)	1	1,200–1,500
9	UPS / Power Backup	APC Back-UPS	1	4,000–7,000

Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Ans:-

 Router Model: TP-Link Archer AX23

- Connect the Router: Connect the broadband modem/ONT to the Wi-Fi router using an Ethernet cable. Switch on both devices.

- Connect to Router: Connect your laptop to the router using Wi-Fi or an Ethernet cable.
- Open Router Settings: Open a web browser and enter 192.168.0.1 or tplinkwifi.net in the address bar.
- Login to Admin Panel: Enter the default login details or create an admin password if requested.
- Set a Unique SSID: Go to Wireless Settings and enter a unique Wi-Fi network name (SSID), such as Bhargav_SOHO_WiFi.
- Enable WPA2 Security: Select WPA2-PSK (AES) as the security option and create a strong Wi-Fi password.
- Change Admin Password: Open System Tools / Administration and change the default admin password to a strong, unique password.
- Save Settings: Click Save or Apply to save all the changes. The router may restart automatically.
- Reconnect Devices: Connect your laptop and Smart TV to the new Wi-Fi network using the updated password.
- Test Internet Connection: Open a website on your laptop and Smart TV to check whether the internet is working properly.

Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Ans:-

IP Address Configuration for SOHO Network

The SOHO network uses static IP addresses for the printer and Smart TV, while laptops receive IP addresses automatically through DHCP.

- Network Configuration:

- Router IP Address: `192.168.1.1`

- Subnet Mask: `255.255.255.0`
- DHCP Range: `192.168.1.100 – 192.168.1.199`

➢ **IP Address Assignment Table**

No.	Device	MAC Address (Example)	IP Address	Configuration
1	Wi-Fi Router	02:11:22:33:44:01	192.168.1.1	Static
2	Printer	02:11:22:33:44:02	192.168.1.10	Static
3	Smart TV	02:11:22:33:44:03	192.168.1.20	Static
4	Laptop	02:11:22:33:44:04	192.168.1.100	DHCP

➢ **Steps to Configure IP Addresses**

1. Configure the Router

- Open the router settings using `192.168.1.1`.
- Log in with the admin password.
- Open DHCP Settings.
  - Enable the DHCP server.
  - Set the DHCP range to `192.168.1.100 – 192.168.1.199`

    - Assign Static IP to Printer

    - Open the printer's Network Settings. ▫ Select Manual IP Configuration. ▫ Enter IP address: `192.168.1.10`.
    - Enter Subnet Mask: `255.255.255.0`. ▫ Enter Gateway: `192.168.1.1`.

2. Assign Static IP to Smart TV

- Open Smart TV Network Settings. ▫ Select Manual IP Settings.
- Enter IP address: `192.168.1.20`.
- Enter Subnet Mask: `255.255.255.0`. ▫ Enter Gateway: `192.168.1.1`.

3. Configure Laptop Using DHCP

a. Open Network Settings on the laptop.
b. Select IP Assignment: Automatic (DHCP).
c. The router will automatically assign an IP address to the laptop.

4. Test the Network

a. Open Command Prompt on the laptop.
b. Type `ipconfig` to check the assigned IP address.
c. Type `ping 192.168.1.10` to test the printer.
d. Type `ping 192.168.1.20` to test the Smart TV.

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Ans:-

Troubleshooting Laptop Internet Connection in SOHO Network

Possible Causes and Solutions Wi-Fi Connection Problem

- Cause: The laptop may be disconnected from Wi-Fi or connected to the wrong network.
- Diagnosis: Check the Wi-Fi icon and verify the SSID.
- Solution: Connect to the correct Wi-Fi network (`Bhargav_SOHO_WiFi`) using the correct password.

Incorrect IP Address

- Cause: The laptop may not receive a valid IP address from the router through DHCP.
- Diagnosis: Open Command Prompt and type `ipconfig` to check the IP address.

- Solution: Run `ipconfig /release` and `ipconfig /renew` to obtain a new IP address automatically.

Router Problem

- Cause: The Wi-Fi router may be frozen or not working properly.
- Diagnosis: Check the router's power and internet indicator lights. Solution: Restart the router and reconnect the laptop to Wi-Fi.

DNS Server Problem

- Cause: Incorrect DNS settings may prevent websites from opening.
- Diagnosis: Open Command Prompt and type `ping 8.8.8.8`. If this works but websites do not open, DNS may be the problem.
- Solution: Set the DNS server to `8.8.8.8` (Google DNS) and try opening the website again.

Internet Service Provider (ISP) Problem Cause: The broadband internet connection may be down.

- Diagnosis: Check whether other devices, such as the Smart TV, can access the internet.
- Solution: Restart the modem/ONT and contact the ISP if the internet is still unavailable.

Outdated or Faulty Wi-Fi Driver Cause: The laptop's Wi-Fi network driver may be outdated or corrupted.

- Diagnosis: Open Device Manager and check Network Adapters for any warning symbols.
- Solution: Update or reinstall the Wi-Fi driver and restart the laptop.

Firewall Blocking Internet Access

- Cause: Windows Firewall or antivirus software may block network traffic.
- Diagnosis: Check Windows Firewall settings and antivirus network protection.
- Solution: Allow the required network connection through the firewall and check internet access again.